

Sleep and Dreams

When we are asleep, it may seem like our brains are quietly resting, but they are actually busy processing and storing information that we have learned throughout the day.

The stages of sleep

During the night, we cycle through different sleep stages, moving from light to deep sleep then to rapid eye movement (REM) sleep. Our brain waves, produced by the electrical activity of neurons in the cortex (see p.42), change in each stage. As sleep becomes deeper, the waves become slower (with lower frequency) and more organized. We repeat this sleep cycle every few hours, but the proportions shift; we have more slow-wave sleep at the start of the night and more REM sleep in the early morning.

A not-so-silent night

There are four distinct stages of sleep, and we pass through each stage several times a night. During light sleep, we are easily woken. It is much harder to wake from deep sleep.

HOW MANY HOURS OF SLEEP DO WE NEED A NIGHT?

Most adults need 7-9 hours of sleep a night, but teenagers and children (especially babies) need more.

Period of wakefulness during night

If woken during REM sleep, we are more likely to remember our dreams

